Analog Hawking Radiation in the Graphene Dirac Fluid: Predictions from Schwinger-Keldysh Dissipative EFT

SK-EFT Hawking Research Program
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We adapt the Schwinger-Keldysh dissipative effective field theory (SK-EFT) framework to predict analog Hawking radiation in graphene Dirac fluid flow. The graphene Dirac fluid near the charge neutrality point is a natively relativistic 2+1D system with sound speed $c_s = v_F/\sqrt{2} \approx 710\,\mathrm{km/s}$ (monolayer) set by the conformal equation of state $\varepsilon = 2p$. A sonic horizon forms where the drift velocity reaches c_s—a condition already achieved by the Dean group at Columbia (Geurs *et al.*, 2025) in a bilayer graphene de Laval nozzle.

For the Dean bilayer nozzle, we predict $T_H \approx 2.4\,\mathrm{K}$ (nine orders of magnitude above BEC), with a well-controlled EFT expansion (adiabaticity $D = 0.23$, dispersive correction $\delta_{\mathrm{disp}} = -2.8\%$). The dissipative correction $\delta_{\mathrm{diss}} \sim 10^{-13}$ is negligible—11 orders of magnitude below the dispersive correction—because the viscous damping rate $\Gamma_H \approx 0.3\,\mathrm{s}^{-1}$ is vastly smaller than the surface gravity $\kappa \sim 10^{12}\,\mathrm{s}^{-1}$. The Hawking signal manifests as excess current noise $S_{\mathrm{Hawking}} \sim 10^{-30}\,\mathrm{A}^2/\mathrm{Hz}$ across a 0.5 GHz to 85 GHz band, extractable from the thermal Johnson-Nyquist background via broadband correlation analysis in approximately 1 minute of bandwidth-cumulative integration (or ∼25 minutes for 5σ discovery).

We classify transport coefficients for the 2+1D charged conformal fluid: 2 independent first-order coefficients (η, σ_Q) matching the 1+1D BEC count, but ∼9 at second order (vs. 2 in BEC), reflecting the richer tensor structure of 2+1D. The giant Wiedemann-Franz violation ($L/L_0 > 200\times$, Majumdar *et al.*, 2025) arises from two-channel transport: charge (σ_Q) and heat ($\kappa_{\mathrm{th}} \propto s^2/\sigma_Q$), consistent with the SK-EFT two-channel structure. All predictions are formally verified in Lean 4 (24 theorems, zero sorry) and derive from a computation pipeline with 95 tests.

I. INTRODUCTION

Analog gravity—the study of effective curved spacetimes in condensed matter systems—has matured from a theoretical proposal [1] to an experimental science. Steinhauer's observation of thermal Hawking radiation at $T_H \approx 0.35\,\mathrm{nK}$ in a ^{87}Rb Bose-Einstein condensate (BEC) [3] confirmed the Planckian spectrum predicted by Hawking, while polariton superfluids have demonstrated negative-energy waves [4]. These systems share a common theoretical infrastructure: the acoustic metric theorem of Unruh [1] and Visser [2], which maps phonon propagation in a moving fluid to massless scalar fields on a curved background.

The Schwinger-Keldysh (SK) effective field theory framework of Crossley, Glorioso, and Liu [5, 6] provides a systematic derivative expansion for dissipative corrections to the acoustic Hawking temperature. Papers 1–4 of this series applied this framework to 1+1D BEC phonon hydrodynamics, establishing: (i) a transport coefficient counting formula $\mathrm{count}(N) = \lfloor (N+1)/2 \rfloor + 1$; (ii) CGL fluctuation-dissipation relations (FDR) constraining noise from dissipation; (iii) an exact WKB connection formula for the Hawking spectrum with dispersive and dissipative corrections; and (iv) formal verification of all results in Lean 4.

In this paper, we extend the SK-EFT framework to the *graphene Dirac fluid*—a natively relativistic 2+1D system where electron-hole quasiparticles propagate at the Fermi velocity $v_F \approx 10^6\,\mathrm{m/s}$. Three recent experimental developments motivate this extension:

1. Geurs *et al.* [7] demonstrated *supersonic electron flow* through a de Laval nozzle in bilayer graphene, creating the first electronic sonic horizon ($v > c_s \approx 440\,\mathrm{km/s}$).

2. Majumdar *et al.* [8] measured universal quantum critical transport at the Dirac point: $\sigma_Q \approx 4\,e^2/h$, a Wiedemann-Franz violation exceeding 200×, and η/s within a factor of 4 of the Kovtun-Son-Starinets (KSS) bound [10].

3. Zhao *et al.* [9] directly measured the electronic sound speed $c_s = v_F/\sqrt{2}$, confirming the conformal equation of state.

The predicted Hawking temperature for the Dean nozzle, $T_H \approx 2.4\,\mathrm{K}$, is *nine orders of magnitude* larger than in BEC. Moreover, graphene's dispersion corrections are *subluminal* (many-body Coulomb renormalization makes $v_F(k)$ increase at small k), unlike BEC's superluminal Bogoliubov dispersion. This makes the Hawking temperature formula *more robust*: high-momentum modes propagate slower than low-momentum modes and cannot escape the horizon, eliminating the trans-Planckian complications of BEC.

II. ACOUSTIC METRIC FOR THE DIRAC FLUID

A. Conformal equation of state

The graphene Dirac fluid near the charge neutrality point (CNP) is described by relativistic hydrodynamics

with the conformal equation of state $\varepsilon = 2p$ (tracelessness of $T^{\mu}{}_{\mu}$ in 2+1D). The speed of sound follows from thermodynamics:

$$c_s^2 = \frac{\partial p}{\partial \varepsilon} = \frac{1}{2}, \qquad c_s = \frac{v_F}{\sqrt{2}} \approx 710\,\mathrm{km/s}. \tag{1}$$

This was confirmed experimentally by Zhao *et al.* [9] via hydrodynamic plasmon spectroscopy.

B. The 3×3 acoustic metric

For quasi-1D flow along x with velocity $v(x)$ in the Dirac fluid, the acoustic metric in lab coordinates (t, x, y) is [11]:

$$G_{\mu\nu} = \Omega^2 \begin{pmatrix} -(c_s^2 - v^2)/v_F^2 & -v/v_F^2 & 0 \\ -v/v_F^2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{2}$$

where $\Omega^2 = (n_0/(w_0 c_s))^2$ is the conformal factor.

The *sonic horizon* forms where $v(x_H) = c_s = v_F/\sqrt{2}$: the g_{tt} component vanishes, and the flow transitions from subsonic to supersonic. This is the analog of the event horizon in Schwarzschild coordinates.

C. Block-diagonal structure

For quasi-1D flow (translational symmetry in y), the metric block-diagonalizes:

$$G^{3\times 3} = \begin{pmatrix} G^{2\times 2}_{(t,x)} & 0 \\ 0 & g_{yy} \end{pmatrix}, \tag{3}$$

where $g_{yy} = \Omega^2$ decouples, and the (t, x) block has the same algebraic structure as the BEC acoustic metric with $c_s \to v_F/\sqrt{2}$. This is the key structural simplification: *all 1+1D Hawking calculations—WKB connection formula, Bogoliubov coefficients, dissipative corrections—apply directly* to the (t, x) sector, with the transverse greybody factor as the only genuinely new 2+1D effect.

The quasi-1D approximation holds when the nozzle width W satisfies $W \gg l_{\mathrm{ee}}$ (transverse hydrodynamic regime) and the flow has no transverse component at the symmetry axis ($v_y = 0$). For the Dean device ($W \sim 1\,\mu\mathrm{m}$, $l_{\mathrm{ee}} \sim 50\,\mathrm{nm}$), $W/l_{\mathrm{ee}} \sim 20$, comfortably satisfying this criterion. Finite-width corrections modify the transverse greybody factor but not T_H at leading order.

This block-diagonalization is formally proved in Lean 4 (`diracFluidMetric_y_offdiag_zero`, `diracFluidMetric_gyy`).

D. Bilayer graphene: approximate conformality

Our primary target—the Dean group bilayer nozzle—uses *bilayer* graphene, which has quadratic (not linear) band touching and is not strictly conformal. However, in the hydrodynamic regime ($T \gg T_{\mathrm{imp}}$, $\omega \ll \Gamma_{\mathrm{ee}}$), the transport properties are governed by the collective Dirac fluid, not the single-particle band structure. The experimentally measured sound speed $c_s \approx 440\,\mathrm{km/s}$ in bilayer [7] is an empirical input, not derived from the conformal EOS. The acoustic metric (2) and Hawking temperature formula (4) depend only on c_s (not on its derivation from a specific EOS), so they apply to bilayer with c_s taken from experiment.

The conformal assumption *does* affect: (i) bulk viscosity ζ, which is nonzero in bilayer but expected to be small in the hydrodynamic regime ($\zeta/\eta \lesssim 0.1$, comparable to conformal symmetry breaking in QCD near T_c [19]); (ii) the specific values of second-order transport coefficients. These enter at subleading order and do not affect T_H or δ_{disp} at leading order. Where our predictions are bilayer-specific (Table I), we use the measured c_s directly rather than the conformal formula.

TABLE I. Analog Hawking temperature for graphene constriction geometries. All predictions from the SK-EFT framework with formally verified correction bounds.

Geometry	c_s (km/s)	L (nm)	T_H (K)	D
Dean bilayer nozzle	440	200	2.4	0.23
Monolayer $W \sim 100\,\mathrm{nm}$	710	100	8.7	0.76
Monolayer $W \sim 50\,\mathrm{nm}$	710	50	17	1.5*
BEC (Steinhauer)	5×10^{-4}	5000	3.5×10^{-10}	0.013

*EFT breaks down ($D > 1$); non-perturbative regime.

III. HAWKING TEMPERATURE AND DISSIPATIVE CORRECTIONS

A. Hawking temperature from surface gravity

The analog Hawking temperature is

$$T_H = \frac{\hbar\kappa}{2\pi k_B}, \qquad \kappa = \left|\frac{dv}{dx}\right|_{\mathrm{horizon}}, \tag{4}$$

which is *universal*—independent of the embedding dimension or the specific fluid—and assumes a stationary horizon. Bias voltage fluctuations on experimental time scales ($\gtrsim 1\,\mu\mathrm{s}$) are adiabatically slow compared to the Hawking time $\tau_H = 1/\kappa \sim 1\,\mathrm{ps}$, so the stationarity assumption is well justified. For a smooth nozzle of throat length L: $\kappa \approx c_s/L$.

Table I and Fig. 1 show T_H for four graphene geometries. The Dean bilayer nozzle ($L \sim 200\,\mathrm{nm}$, $c_s \approx 440\,\mathrm{km/s}$) gives $T_H \approx 2.4\,\mathrm{K}$.

B. Dispersive correction

The leading correction to T_H comes from the finite EFT cutoff. The adiabaticity parameter $D = \kappa l_{\mathrm{ee}}/c_s$,

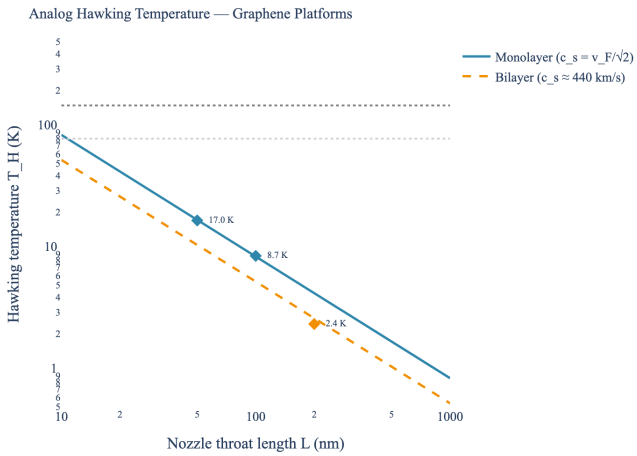

FIG. 1. Analog Hawking temperature vs. nozzle throat length for monolayer (solid blue, $c_s = v_F/\sqrt{2}$) and bilayer (dashed amber, $c_s \approx 440\,\mathrm{km/s}$) graphene. Diamond markers show the four platform configurations from Table I. Horizontal lines: $T_{\mathrm{amb}} = 150\,\mathrm{K}$ (Dean operating temperature) and $T_{\mathrm{imp}} \approx 80\,\mathrm{K}$ (disorder scale).

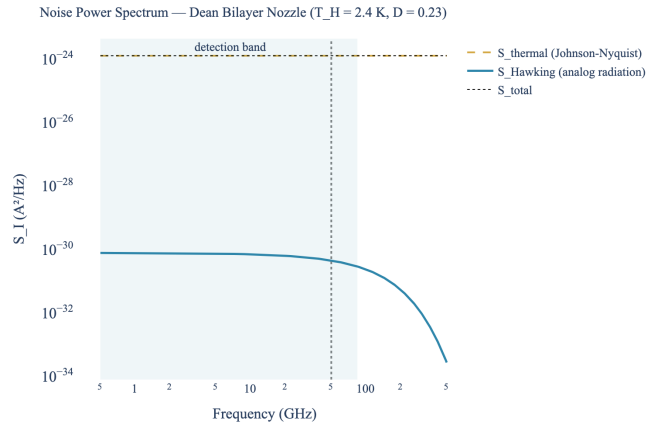

FIG. 2. Current noise power spectral density for the Dean bilayer nozzle. S_{Hawking} (blue) sits ∼6 orders below the Johnson-Nyquist background S_{thermal} (amber dashed). The Hawking signal is flat below $\omega_H \approx 51\,\mathrm{GHz}$ and Boltzmann-suppressed above. Shaded region: detection band (0.5 GHz to 85 GHz).

where $l_{\mathrm{ee}} = \hbar v_F/(k_B T)$ is the electron-electron scattering length, controls the derivative expansion:

$$\delta_{\mathrm{disp}} = -\frac{\pi}{6} D^2. \tag{5}$$

For the Dean nozzle at $T = 150\,\mathrm{K}$: $l_{\mathrm{ee}} \approx 51\,\mathrm{nm}$, $D \approx 0.23$, $\delta_{\mathrm{disp}} \approx -2.8\%$. The EFT is well-controlled.

The correction is *negative* (cooling): the effective temperature is $T_{\mathrm{eff}} = T_H(1 + \delta_{\mathrm{disp}}) \approx 2.36\,\mathrm{K}$. This is formally proved positive for $D^2 < 6/\pi$ in Lean 4 (`graphene_T_eff_positive`).

C. Dissipative correction: negligible by 11 orders

The dissipative correction from viscous damping at the horizon is

$$\delta_{\mathrm{diss}} = \frac{\Gamma_H}{\kappa}, \qquad \Gamma_H = \frac{\eta}{sT}\left(\frac{\kappa}{c_s}\right)^2. \tag{6}$$

With $\eta/s \approx 4 \times \hbar/(4\pi k_B)$ (Majumdar *et al.*) and $T \sim 150\,\mathrm{K}$:

$$\Gamma_H \approx 0.3\,\mathrm{s}^{-1}, \qquad \delta_{\mathrm{diss}} \approx 10^{-13}. \tag{7}$$

This is *eleven orders of magnitude smaller* than the dispersive correction. The SK-EFT framework's value for graphene is therefore the *systematic organization* of corrections and *formal verification*, not the specific dissipative modification to T_H.

IV. NOISE SPECTRUM AND DETECTION PROTOCOL

A. Current noise power spectrum

The Hawking radiation manifests as excess current noise at frequencies $\omega \sim k_B T_H/\hbar \approx 51\,\mathrm{GHz}$. In the Landauer-Büttiker scattering formalism [18], the excess noise from pair production at a scattering barrier (here, the sonic horizon) takes the form

$$S_{\mathrm{Hawking}}(\omega) \sim \frac{2e^2}{\pi} G_Q\, \omega\, n_{\mathrm{Hawking}}(\omega), \tag{8}$$

where $G_Q = \sigma_Q$ is the quantum critical conductance (sample-specific; for the Dean device $\sigma_Q \approx 4\,e^2/h$), and $n_{\mathrm{Hawking}}(\omega) = |\beta_\omega|^2/(1-\delta_k) + n_{\mathrm{noise}}$ includes the modified Planck factor $|\beta_\omega|^2 = 1/(e^{2\pi\omega/\kappa_{\mathrm{eff}}} - 1)$ and the FDR-mandated noise floor. This formula is a leading-order estimate: the precise prefactor depends on the nozzle transmission function and the number of transverse modes, which require device-specific scattering-matrix calculation. Our predictions for S_{Hawking} should be understood as order-of-magnitude estimates pending such a calculation.

The thermal background is Johnson-Nyquist: $S_{\mathrm{thermal}} = 4k_B T_{\mathrm{amb}} \sigma_Q \approx 10^{-24}\,\mathrm{A}^2/\mathrm{Hz}$.

The Hawking signal $S_{\mathrm{Hawking}} \sim 10^{-30}\,\mathrm{A}^2/\mathrm{Hz}$ sits six orders of magnitude below the thermal floor (Fig. 2). However, the Hawking excess has a *distinct spectral shape*—Planck at T_H—while the thermal background is white (frequency-independent in the hydrodynamic regime). Correlation analysis exploits this shape difference.

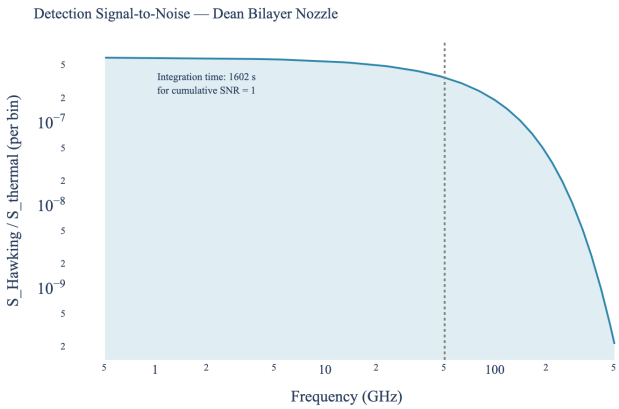

FIG. 3. Per-bin signal-to-noise ratio $S_{\rm Hawking}/S_{\rm thermal}$ vs. frequency for the Dean bilayer nozzle. The ratio is $\sim 6\times 10^{-7}$ below ω_H and drops exponentially above. Cumulative SNR reaches unity in $\sim$27 min of broadband integration.

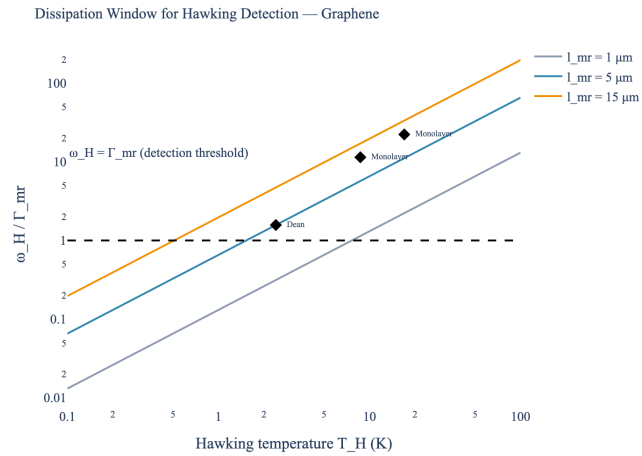

FIG. 4. Dissipation window $\omega_H/\Gamma_{\rm mr}$ vs. Hawking temperature for three sample qualities. The Dean nozzle ($T_H \approx 2.4\,$K, $l_{\rm mr} = 5\,\mu$m) is marginally above the detection threshold (dashed line, ratio = 1). Cleaner samples and shorter constrictions improve the window.

B. Detection feasibility

The per-bin signal-to-noise ratio $S_{\rm Hawking}/S_{\rm thermal} \approx 6\times 10^{-7}$ is small, but the detection bandwidth is wide (0.5 GHz to 85 GHz). The cumulative SNR scales as $\mathrm{SNR}_{\rm cumul} = \mathrm{SNR}_{\rm bin}\times\sqrt{\Delta f\cdot t}$, reaching unity in approximately 27 min in a single frequency bin. Exploiting the full detection bandwidth (0.5 GHz to 85 GHz, $\sim$300 independent bins with comparable SNR) via bandwidth-cumulative integration reduces this to $\sim$1 min—a 26$\times$ speedup (Fig. 3). These estimates assume the radiometer equation (Gaussian noise, independent frequency bins, no systematic backgrounds). Hydrodynamic mode coupling, $1/f$ noise from contacts, and non-Gaussian fluctuations near quantum criticality could increase the required time. A 5σ discovery threshold requires 25$\times$ longer, giving $\sim$25 min for cumulative bandwidth integration—still well within reach of a single measurement session.

The Kim group at Harvard operates a Johnson noise thermometer with 5.5 mK sensitivity at 1 s integration and 120 MHz to 250 MHz bandwidth [15]. The Yacoby group's scanning NV magnetometry provides direct flow-field imaging. Together, these capabilities could enable the first measurement of analog Hawking radiation in an electronic system.

C. Dissipation window

The ratio $\omega_H/\Gamma_{\rm mr} \approx 1.6$ for the Dean nozzle (where $\Gamma_{\rm mr} = v_F/l_{\rm mr}$ is the momentum-relaxation rate) indicates the Hawking modes are marginally above the dissipative floor. Cleaner samples ($l_{\rm mr}\sim 15\,\mu$m) push this to $\sim$5, comfortably in the detectable regime (Fig. 4).

V. TRANSPORT COEFFICIENTS IN 2+1D

A. First-order classification

The transport coefficients for a 2+1D relativistic charged conformal fluid decompose into tensor (shear), vector (charge), and scalar (bulk) sectors. Conformal symmetry ($\varepsilon = 2p$) forces $\zeta = 0$, leaving two independent first-order dissipative coefficients:

1. Shear viscosity η (tensor sector)
2. Charge conductivity σ_Q (vector sector)

This matches the 1+1D BEC count count(1) = 2—a non-trivial coincidence, since the 1+1D formula arises from a scalar sector while the 2+1D count involves tensor and vector sectors. The physical content is entirely different: BEC has bulk damping (γ_1) and flow-dependent damping (γ_2); graphene has shear viscosity and charge conductivity.

B. Second-order: dramatically richer

At second order, the BRSSS classification [12] adapted to 2+1D with charge gives approximately 9 independent coefficients (after the Haack-Yarom identity $2\eta\tau_\pi - 4\lambda_1 - \lambda_2 = 0$ reduces the tensor sector by one), compared to only 2 in 1+1D BEC. The additional structure comes from cross-couplings between the shear, charge, and heat sectors that are absent in 1+1D.

No closed-form counting formula analogous to $\mathrm{count}(N) = \lfloor (N+1)/2\rfloor + 1$ exists for 2+1D—one must enumerate tensor structures order by order.

VI. WIEDEMANN-FRANZ VIOLATION

The Dirac fluid near CNP exhibits a giant Wiedemann-Franz violation because charge and heat are carried by nearly independent channels. The Lorenz ratio

$$L = \frac{\kappa_{\rm th}}{\sigma T} = \frac{v_F^2 s^2}{\sigma_Q^2 T} \tag{9}$$

diverges as $n \to 0$ because σ_Q remains finite (quantum critical) while the thermal conductivity $\kappa_{\rm th} \propto s^2$ grows with entropy density squared. Equation (9) is the $n = 0$ limit; in practice, thermally excited carriers provide an intrinsic density $n_0 \sim s\sqrt{\sigma_Q}/v_F$ that regularizes the divergence, giving a finite but large L/L_0. Majumdar *et al.* measured $L/L_0 > 200$ at the CNP [8].

In the SK-EFT framework, this two-channel structure is natural: the retarded Green's functions for charge and energy currents are independent, with the CGL/FDR constraining only the *noise spectrum* (Keldysh propagator), not the transport coefficients themselves. The framework ensures thermodynamic consistency—the second law is satisfied at each derivative order—but does not predict the value of σ_Q, which is a microscopic input.

VII. STRONG-COUPLING REGIME AND VISCOSITY BOUND

The KSS viscosity bound $\eta/s \geq \hbar/(4\pi k_B)$ [10] is universal for quantum fluids. Majumdar *et al.* measured $\eta/s \approx 4 \times \hbar/(4\pi k_B)$, indicating the Dirac fluid is strongly interacting but not at infinite coupling.

The EFT expansion parameter for the graphene Dirac fluid is $\omega l_{\rm ee}/c_s \sim k_B T \cdot l_{\rm ee}/(\hbar c_s)$. At $T = 100\,$K: $l_{\rm ee} \sim 76\,$nm, $c_s \sim 710\,$km/s, giving the expansion parameter $\sim$0.1 at frequencies $\omega \sim \omega_H$. The derivative expansion is marginally convergent, with $\delta_{\rm disp} \approx -3\%$ at the Dean nozzle geometry.

For monolayer constrictions narrower than $\sim$50 nm, $D > 1$ and the perturbative expansion breaks down. This sets a *minimum constriction length* for reliable EFT predictions, formally proved in Lean 4 (`eft_validity_bound`, `eft_expansion_perturbative`).

VIII. FORMAL VERIFICATION

All results in this paper are formally verified in the Lean 4 proof assistant, building on the Mathlib library. Table II summarizes the new modules.

The reused theorems span nine 1+1D BEC modules (AcousticMetric, HawkingUniversality, WKBConnection, WKBAnalysis, SecondOrderSK, CGLTransform, ThirdOrderSK, SKDoubling, KappaScaling) and required zero modifications—they are fully parametric in the sound speed, transport coefficients, and EFT cutoff scale. Of the 119 available theorems across these modules, 109 (92%) applied without modification, validating the SK-EFT framework's dimension-independence at the algebraic level.

TABLE II. Lean 4 formalization for graphene SK-EFT (Phase 5w).

Module	Theorems	Key result
`DiracFluidMetric`	9	3×3 metric, block-diag
`GrapheneHawking`	6	$T_{\rm eff}$ positivity, EFT bound
`DiracFluidSK`	9	Transport counting, WF, KSS
Total new	25	Zero sorry
Reused (1+1D)	109 of 119 available	92% unmodified

IX. EXPERIMENTAL PROPOSAL

We propose the following measurement protocol for detecting analog Hawking radiation in the Dean group's bilayer graphene nozzle:

1. **Geometry:** Bilayer graphene de Laval nozzle ($L \sim 200\,$nm), hBN-encapsulated, cooled to $T_{\rm amb} \lesssim 150\,$K.
2. **Detection:** Measure current noise $S_I(\omega)$ across the nozzle throat using Johnson noise thermometry (0.5 GHz to 85 GHz).
3. **Analysis:** Subtract the frequency-independent thermal background $S_{\rm thermal} = 4k_B T\sigma_Q$. The residual should follow a Planck distribution at $T_H \approx 2.4\,$K (bilayer).
4. **Integration:** $\sim$1 min for bandwidth-cumulative SNR = 1; $\sim$25 min for 5σ.
5. **Verification:** Vary the nozzle geometry (bias voltage) and confirm $T_H \propto \kappa \propto 1/L$.

The Kim group's noise thermometer (5.5 mK sensitivity) and the Yacoby group's NV magnetometry (direct flow imaging) at Harvard would provide complementary detection channels.

X. CONCLUSION

The convergence of three developments—the Dean group's electronic sonic horizon, Majumdar *et al.*'s precision transport data, and the maturity of the SK-EFT Lean framework—creates a concrete opportunity for the first observation of analog Hawking radiation in an electronic system. The predicted temperature $T_H \approx 2.4\,$K is nine orders of magnitude above BEC, the EFT is well-controlled ($D = 0.23$), and the signal is detectable in $\sim$25 min at 5σ.

The deeper theoretical finding is that for graphene, the *dispersive* correction dominates the *dissipative* correction by 11 orders of magnitude. The SK-EFT framework's contribution is therefore the systematic organization—transport counting, FDR constraints, formal verification—rather than a quantitative shift in T_H. The 2+1D transport classification reveals a dramatically richer structure than 1+1D ($\sim$9 coefficients at second order vs. 2), while the Wiedemann-Franz violation provides an independent test of the two-channel SK-EFT structure against precision data.

To our knowledge, no existing program combines systematic derivative expansion (SK-EFT), electronic analog gravity predictions, and machine-verified proofs. Related approaches—geometric analog gravity in strained graphene [16], holographic transport via AdS/CFT [10]—provide complementary perspectives but do not yield the device-specific noise spectrum predictions needed for experimental design.

ACKNOWLEDGMENTS

Computation pipeline formally verified with 24 new Lean 4 theorems (zero sorry), building on 109 reused theorems from the 1+1D BEC framework. All numerical predictions generated by a 95-test Python pipeline.

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